

Control work 5, and work on the mistakes

Expressions, powers, monomials and polynomials in one paper — Quarter II closed.

LESSON 79–80

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 5

S8 2 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Apply the laws of indices without confusing them with the laws of arithmetic.
- Add, subtract, multiply and divide polynomials accurately under time.
- Factorise by the common factor and by grouping.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 152–224
Cambridge	Stage 8, pp. 12–30
Workbook	Control paper M5

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Simplify $a^5 \cdot a^3$, $a^8 \div a^3$, $(a^2)^4$ and a^0	4	L63–64
2	Multiply the monomials $3x^2y$ and $-4xy^3$	3	L66–67
3	Collect $5a^2 - 3a + 2 - 2a^2 + 7a - 5$	4	L69–70
4	Expand $(x + 4)(x - 3)$ and $-3a(a^2 + 2a - 5)$	5	L73–74
5	Divide $(10x^3y - 15x^2y^2) \div 5x^2y$	4	L75–76
6	Factorise $12a^2b + 8ab^2 - 4ab$ and $2x^2 + 6x + ax + 3a$	5	L77–78

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for $a^0 = 1$ alone; Q2 one for the sign; Q4 two for the four products of the binomial pair and one for the sign of the last term; Q6 one for the -1 left by the third term and two for the common bracket in the grouping.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
indices multiplied	$a^5 \cdot a^3 = a^{15}$	a^8 — add them
zero index misread	$a^0 = 0$	$a^0 = 1$
sign of a product	$3x^2y \cdot (-4xy^3) = 12x^3y^4$	$-12x^3y^4$
unlike terms collected	$5a^2 - 3a = 2a$	they are not like
only two products	$(x + 4)(x - 3) = x^2 - 12$	all four, then collect
last term of a division dropped	$2x$	$2x - 3y$
the 1 lost when factorising	$4ab(3a + 2b)$	$4ab(3a + 2b - 1)$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Quarter II as one map

BLOCK	THE SENTENCE
expressions	a letter stands for a number, so the laws of arithmetic still hold
powers	an index counts the factors, so multiplying adds the indices
monomials	one product of numbers and letters, with a degree
polynomials	a sum of monomials; only like terms may be collected
multiplication	every term of the first by every term of the second
division	term by term, and only when the divisor is a common factor
factorising	the common factor outside, or grouping in pairs

LOOKING FORWARD

Quarter III opens with geometry again — parallel lines, the angle sum of a triangle and right-angled triangles — and returns to algebra with the abridged multiplication formulae, which are the products of Q4 learned by heart.

02 Worked examples

EXAMPLE 1 Model answer, Q1: the four index tasks.

① $a^5 \cdot a^3 = a^8$
Add the indices.

② $a^8 \div a^3 = a^5$
Subtract them.

③ $(a^2)^4 = a^8$
Multiply them.

④ $a^0 = 1$
For every $a \neq 0$.

ANSWER

$a^8, a^5, a^8, 1$

EXAMPLE 2 Model answer, Q4: expand $(x + 4)(x - 3)$.

$$1 \quad x \cdot x = x^2$$

$$2 \quad x \cdot (-3) = -3x \text{ and } 4 \cdot x = 4x.$$

$$3 \quad 4 \cdot (-3) = -12$$

The sign is the mark.

$$4 \quad = x^2 + x - 12$$

ANSWER

$$x^2 + x - 12$$

EXAMPLE 3 Model answer, Q6: factorise $2x^2 + 6x + ax + 3a$.

$$1 \quad (2x^2 + 6x) + (ax + 3a)$$

$$2 \quad 2x(x + 3) + a(x + 3)$$

$$3 \quad = (x + 3)(2x + a)$$

$$4 \quad \text{Check by expanding: } 2x^2 + ax + 6x + 3a \checkmark$$

ANSWER

$$(x + 3)(2x + a)$$

03 Interactive model

Hand back Q4 and Q6 together and ask which is which: expanding and factorising are one skill in two directions, and seeing them side by side fixes it.

Quarter II in twelve questions

 $a^5 \cdot a^3$ equals:

a^{15}

a^8

a^2

$2a^8$

 $a^8 \div a^3$ equals:

a^5

a^{11}

a^2

a^{24}

 $(a^2)^4$ equals:

a^6

a^8

a^{16}

$2a^4$

a^0 equals:

0

1

a

undefined

The degree of $3x^2y$ is:

2

3

5

1

$3x^2y \cdot (-4xy^3)$ equals:

$12x^3y^4$

$-12x^3y^4$

$-12x^2y^3$

$-7x^3y^4$

Like terms differ only in:

the letters

the coefficient

the degree

the sign of the letter

$5a^2 - 3a + 2 - 2a^2 + 7a - 5$ equals:

$(x + 4)(x - 3)$ equals:

$(8a^3 - 4a^2) \div 4a^2$ equals:

$12a^2b + 8ab^2 - 4ab$ factorises to:

A factorisation is checked by:

dividing

expanding

substituting

nothing

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Power	Daraja	Степень
Monomial	Bir had	Одночлен
Polynomial	Ko'phad	Многочлен
Like terms	O'xshash hadlar	Подобные члены
To expand	Qavsni ochish	Раскрыть скобки
To factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 is answered with:

the laws of arithmetic

the laws of indices

a calculator

estimation

The sign in Q2 comes from:

the letters

one negative factor

the degree

the coefficients only

In Q3 the terms $5a^2$ and $-3a$ are:

like

not like

equal

opposite

Q4 needs how many products for the binomial pair?

2

3

4

6

Q5 is wrong if:

the first term is divided

the last term is dropped

the check is done

the answer is a polynomial

The commonest slip in Q6 is:

the wrong HCF

the lost 1

too many brackets

expanding

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $a^5 \cdot a^3$

ANSWER a^8

2. $a^8 \div a^3$

ANSWER a^5

3. $(a^2)^4$

ANSWER a^8

4. a^0 , for $a \neq 0$

ANSWER 1

5. $3x^2y \cdot (-4xy^3)$

ANSWER $-12x^3y^4$

6. $5a^2 - 2a^2$

ANSWER $3a^2$

7. $(x^2 + x) \div x$

ANSWER $x + 1$ **Medium · the standard question**

7 PROBLEMS

1. $5a^2 - 3a + 2 - 2a^2 + 7a - 5$

ANSWER $3a^2 + 4a - 3$

2. $(x + 4)(x - 3)$

ANSWER $x^2 + x - 12$

3. $-3a(a^2 + 2a - 5)$

ANSWER $-3a^3 - 6a^2 + 15a$

4. $(10x^3y - 15x^2y^2) \div 5x^2y$

ANSWER $2x - 3y$

5. $12a^2b + 8ab^2 - 4ab$

ANSWER $4ab(3a + 2b - 1)$

6. $2x^2 + 6x + ax + 3a$

ANSWER $(x + 3)(2x + a)$

7. The degree of $3x^2y$

ANSWER 3

Hard · stretch and reason

7 PROBLEMS

1. $(2x - 1)(x^2 + x - 3)$

ANSWER $2x^3 + x^2 - 7x + 3$

2. $(15x^4y^2 - 10x^3y^3 + 5x^2y) \div 5x^2y$

ANSWER $3x^2y - 2xy^2 + 1$

3. $x(a - b) - y(b - a)$

ANSWER $(a - b)(x + y)$

4. Simplify $\frac{6x^2 + 9x}{3x}$

ANSWER $2x + 3$

5. Expand and collect $(a + b)^2 - (a - b)^2$

ANSWER $4ab$

6. $48 \cdot 37 + 48 \cdot 63$

ANSWER 4800

7. Why is $n^2 + n$ always even?

ANSWER $n(n + 1)$ — one factor is even**07 Homework****Homework — 5 tasks**

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Write out the four laws of indices with one example each.
3. Expand $(2x - 5)(x + 3)$ and check by substituting $x = 1$.
4. Factorise $9m^2n + 12mn^2 - 3mn$ and expand your answer back.

5. Write one sentence for each of the seven blocks on the Quarter II map.